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ITAI 3377

A03 Case Study Analysis on Edge-Computing Video Analytics

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A. Introduction and Objectives

The Liverpool Smart Pedestrians project was established to develop a non-intrusive method for collecting urban mobility data to directly inform future city planning initiatives (Barthélemy et al., 2019). The primary urban planning challenge addressed was the rapid expansion of the Liverpool city center, which was projected to experience an influx of approximately 30,000 additional pedestrians every day due to extensive CBD redevelopment (Barthélemy et al., 2019). By implementing a real-time monitoring system, the project sought to provide city officials with the granular data necessary to manage this growth sustainably while balancing social and infrastructural pressures (Barthélemy et al., 2019).

B. Methodology

The project utilized a methodology centered on the creation of an intelligent visual sensor that leverages the edge-computing paradigm to process video data locally at the source (Barthélemy et al., 2019). Early community workshops helped define essential requirements, such as the need for multi-modal detection of diverse road users—including pedestrians, cyclists, and motor vehicles—while ensuring strict privacy compliance (Barthélemy et al., 2019). To remain cost-effective and scalable, the system was designed with the constraint of retrofitting existing CCTV infrastructure, ensuring that new sensors could be added or relocated without requiring entirely new network foundations (Barthélemy et al., 2019).

C. Technology and Implementation

The hardware implementation relied on the NVIDIA Jetson TX2, an ARM-based computing device chosen for its specialized ability to accelerate neural network computations while remaining power-efficient (Barthélemy et al., 2019). For software, the project selected the YOLO V3 algorithm, which provided a robust balance between processing speed and the accuracy required to detect objects at multiple scales (Barthélemy et al., 2019). This implementation perfectly aligns with the edge-computing principles discussed in Module 2, specifically the "edge layer" function of performing localized, low-latency processing to reduce bandwidth usage and enhance data security by only transmitting anonymous metadata (University of Wollongong, 2025a).

D. Validation and Performance

Validation experiments conducted using the Oxford Town Center Dataset revealed that the smart sensor achieved a mean accuracy of 69% and a median processing speed of 19.57 frames per second (Barthélemy et al., 2019). While these results were considered satisfactory for identifying traffic trends, the analysis identified a critical system bottleneck where the SORT tracking algorithm was limited by the CPU, causing frame rates to drop as crowd density increased (Barthélemy et al., 2019). Despite these issues, system utilization remained stable during real-world testing, with the GPU handling most detection tasks while outbound network traffic was significantly lower than inbound data due to the efficiency of edge processing (Barthélemy et al., 2019).

E. Real-World Applications

The sensor's versatility was demonstrated through two distinct applications: an indoor deployment that successfully tracked building occupancy and movement during a fire evacuation, and an outdoor deployment in Liverpool that captured a week's worth of city traffic patterns (Barthélemy et al., 2019). These experiments highlighted the impact of sensor technologies in revealing the "circadian rhythms" of a city, allowing urban planners to visualize pedestrian flows and detect anomalies like unauthorized line crossings (Barthélemy et al., 2019). As covered in Module 3, the use of edge gateways in these scenarios allows for efficient data aggregation and pre-processing, transforming raw visual data into actionable insights for safer and more efficient urban environments (University of Wollongong, 2025b).

F. Challenges and Future Work

The main challenges faced during implementation included software bottlenecks in the tracking algorithm and reduced accuracy due to occlusions in large crowds, for which the authors proposed migrating to more powerful platforms like NVIDIA Xavier (Barthélemy et al., 2019). Since the project's conclusion in 2019, technological advancements such as the NVIDIA Orin series now provide up to 275 TOPS of AI performance, vastly exceeding the capabilities of the original TX2 (NVIDIA Developer, 2025). Furthermore, the development of end-to-end, NMS-free vision models like YOLO26 and high-reliability communication protocols such as 5G

RedCap offer significantly lower latency and better support for the high device density required for modern smart cities (Ultralytics, 2026; Com4, 2025).

G. Personal Evaluation

In my evaluation, the Liverpool project was a pioneering success in demonstrating that high-quality urban analytics and citizen privacy are not mutually exclusive when using edge-computing. The methodology of leveraging existing CCTV infrastructure is particularly impressive as it offers a sustainable model for resource-constrained municipalities. To further enhance the system, I would recommend integrating a "Zero Trust" security architecture as mentioned in Module 2 to protect edge devices from evolving cyber threats (University of Wollongong, 2025a). Additionally, adopting modern transformer-based vision architectures could help overcome the occlusion challenges identified in the original study, leading to even more precise traffic modeling.

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